

Boss Baby: Foamy Business

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Figure 1: Boss Baby Foam. Left: close-up foam piles; center: foam-character interaction; right: large-scale explosive foam.

ABSTRACT

In a world run by babies, the most potent weapon from *The Boss Baby: Family Business* was, appropriately, the loveable, children's craft - foam. Yet needing to show foam as both a friendly, intimate plaything and a powerful, massively dangerous adversary, we extended our proprietary Material Point Method solver to represent close up interactions retaining the mass of the frothy material as well as creating large scale, explosive volumetric setups. To retain a consistent look, we created a robust geometric surface and textural shader that was able to hold its form whether fitting in a handheld glass or expanding at speeds and scales to fill an entire courtyard.

CCS CONCEPTS

• Computing methodologies → Physical simulation; Simulation by animation; Procedural animation.

KEYWORDS

Foam effects, procedural animation, MPM

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1 MPM SOLVER FOR DETAILED FOAM

One of the primary uses for our MPM foam solver was close up and character interaction shots. The precise mass-preserving simulation and the rolling motion and speed variations within the simulation allowed outside forces to manipulate the foam without degrading and losing its foamy consistency, as it would in standard dynamic solvers. Various preset settings allowed the quick generation of wedged iterations to select and establish a look early in the development timeline.

We modeled parts of the foam as a viscoelastic material, following the constitutive model listed on [Yue et al. 2015]. There were 8 different material parameters that a user could change; the primary four being: density, bulk modulus, shear modulus and yield. The foam material parameters were stored on a per-particle basis; this approach provides more artistic control over specific shapes and dynamics without sacrificing the look of the surface of the foam.

To implement the paper in a production form, we brought down the initial baseline substep count of around 4000 [Yue et al. 2015] into a range of 300-600 substeps per frame. The primary tradeoff was to limit the strain that the material could sustain. This allowed us to simulate 1 million points with a voxel size of 2.5 cm, resulting in a simulation time of 1.5 minutes per frame. The simulation was run on two 28-cores, 2.70GHz Intel Xeon CPU processors.

2 SCALING EXPLOSIVE FOAM

The "babies" used Mentos and soda as a playtime table-top activity as well as a lightning-fast weapon that chased down a villain and exploded into a giant atrium over multiple shots and angles. There was no one-size-fits-all solution for this, but we were able to generate setups to match all the necessary variations.

2.1 VDB based foam expansion

To handle the production shots of the explosive foam, we modeled and animated base shapes to block in overall size and expanding motion so that upstream departments could properly set up

the character acting. These shapes became inputs into a custom VDB [Museth 2013] based simulation with an average simulation time of about 1 min/frame. Shots varied from resolutions of 5 million voxels to 9 million voxels. At each timestep, we expanded the shapes based on the signed distance field gradients of the VDB and added noise to enhance the feeling of organic foam shapes. Custom attributes were created to control the rate of expansion and aggressiveness of the noise over the course of each timestep. This gave us direct control to match the action of the characters, and choreograph the "acting" of the foam, in correspondence to the characters' facial and body reactions.

2.2 Multiple simulations

Attributes such as curvature and expansion velocity from the base simulation were used to design several second level flip and vellum simulations. Fine tuning the viscosity and surface tension helped maintain the small-scale, yet complex feeling of the foam, even as it was rapidly expanding during an explosion. We used dual rest fields and vorticity to add further displacement and extra detail to the surfaces. Our bubble generation technique kept both large and small scale shots within a similar range of bubble quantity to prevent the need for vastly larger numbers of bubbles in wide shots, as they would be in the real world, from linearly scaling up geometry and processing time. In close up shots, the average particle count was around 1.5 million, while the average voxel count was 200 million. In wider shots, the average particle count was 5 million and the average voxel count was 400 million, giving us a world space size increase of 10-15 times with only 2 times more processing.

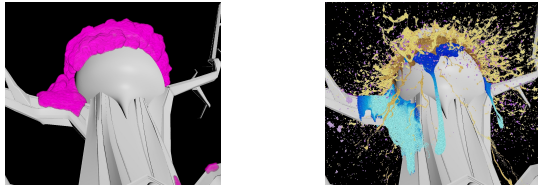


Figure 2: Base foam expansion; multiple simulation layers.

2.3 Tiny Bubbles Generation

We used two different methods to generate bubbles. (1) In close up shots, a custom rigid body based solver was used to manage the bubble clusters covering the surface of the foam. Using the combined fields from the previous simulations, spheres were sourced and constrained to the surfaces as they were advected on each simulation frame. Particle dynamics were used to manage the size distribution, attraction forces, and birthing of the bubbles throughout the simulation. The silhouette of the foam was important, as it projected the foam's intention on a quick read by the audience. To better art-direct this, surface tension was modeled as a post process by using curvature to blend bubbles with the surface of the foam. (2) In the wider shots, we used volume gradients to push a portion of particles from previous flip simulations towards the surface of the foam for better visual coverage. This sacrificed the visual depth detail that would not be noticed at that scale anyway. Fields from the simulations were used to steer the particles in motion. The

points were procedurally relaxed to prevent bubble geometry intersections, then camera frustum culling reduced the bubble count by an average of 60%.

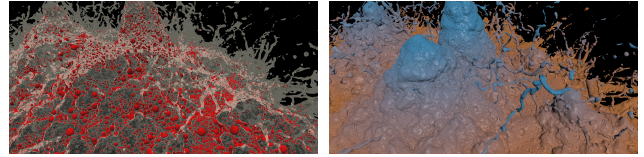


Figure 3: Relaxed surface bubbles; mesh subtracting bubbles.

3 FOAM RENDERING

When we meshed the foam surface and simulations together to render, the bubbles were subtracted from the output mesh to avoid the darkness of self-shadowing and allow the extension of subsurface scattered light. Initial attempts using geometric operations proved prohibitively slow for the large scale shots, but switching to purely VDB combination methods vastly increased memory efficiency while retaining the desired detail reflected in the close-up shots.

In our proprietary render engine, MoonRay, we blended a volumetric shader with a refractive shader to render the base foam mesh. This allowed us to preserve and create naturalistic light in the foam volume through anisotropic scattering while supporting the impression of reflections on the foam surface. For exterior bubbles, we used a refractive water shader for a liquid sheen. However, there was a shadowing issue with the foam mesh, so we rendered an additional bubble layer with reduced transparency just for casting shadows. This separation, along with increasing the ray march limit, reduced harsh shadowing and allowed the surface to feel like it contained even more microscopic bubbles than were physically there.

4 CONCLUSION

The Boss Baby: Family Business extended the application of foam unseen thus far, pushing us to develop a flexible and controllable setup to address all its varying uses. Our MPM techniques identified and applied the balance between theoretical measurements and production usage, creating discrete material settings for quick artistic development and running as fast as 1.5 minutes per frame, while reducing the substep requirement by more than 10 times with strain limitations. Multiple layers of interacting simulations gave us explicit control to previsualize and animate the foam together with the character actors. Our blended shading on the bubble surface provided a dynamic surface appearance that reduced the geometric footprint by up to 8 times while retaining depth and complexity. Most importantly, we were able to carry on the youthfully adventurous quality that defines *The Boss Baby* franchise.

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